Seven squares from three numbers

Problèmes ouverts — A. Dujella · 2026

Problem 5.17 (Dujella, 2026). Are there infinitely many triples of distinct non-zero rationals (a, b, c) such that $a+1$, $b+1$, $c+1$, $ab+1$, $ac+1$, $bc+1$, and $abc+1$ are all squares ?

Remark. Some triples with the given property are $(8, \frac{15}{49}, \frac{1320}{49})$, $(8, \frac{312}{529}, \frac{495}{529})$, $(\frac{25}{144}, -\frac{143}{144}, \frac{1792}{3249})$. By the recent paper Dujella, A., Szalay, L. : Four squares from three numbers. Preprint (2025). arXiv : 2506.14013, there are infinitely many triples (a, b, c) of integers greater than 1 such that $ab+1$, $ac+1$, $bc+1$, and $abc+1$ are all squares.

Problem solved in the recent paper Dujella, A., Kazalicki, M., Petričević, V. : Seven squares from three numbers. Preprint (2026). arXiv :2604.08729. The solution is based on the fact that the quartic $3u^4 - 9u^2 + 7 = z^2$ has infinitely many rational points. This gives an infinite sequence of triples with the given property, starting with $(8, \frac{312}{529}, \frac{495}{529})$, $(\frac{312}{529}, -\frac{152880}{165649}, -\frac{78374557}{87628321})$, $(-\frac{152880}{165649}, \frac{724255280}{736742449}, -\frac{63009087694401}{122040649934401})$, $(\frac{724255280}{736742449}, \frac{24490915482072}{12448992625969}, -\frac{4510665894525110607837}{9171701314839342058081})$, $(\frac{24490915482072}{12448992625969}, -\frac{2539564321528123032}{50545459072823294 41}, \frac{1426184240434933134595097481969 5}{6292400472737950798798594985332 9})$. In the same paper, it is proved that there does not exist a triple of non-zero integers with the same property.